

Richard Álvarez

Experience

AI Solutions Engineer

Sep 2025 – Present

Layer 7 Systems | Independent Contractor

Remote

- Built a stateful TypeScript, Bun, and Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph and Teams for structured service desk intake, ticket actions, and technician routing, then operated it in production for several months.
- Engineered caller and company resolution, contact-safe ticket access, backend authorization gates, deterministic transfer eligibility, post-call workflows, and human fallback to prevent unsafe writes and cross-client data exposure.
- Built an authenticated React operations dashboard end to end and expanded and maintained a 750+ test regression suite covering authorization, validation, transfers, external API behavior, alerts, and failure recovery.
- Deployed Docker workloads through GitHub Actions across staging and production, then diagnosed duplicate tickets, stale identity state, API drift, retry and dead-letter queue failures, transfer defects, and CI and VPS incidents.
- Built a workflow-analysis pipeline that reviewed 60 recordings from 14 users and identified 176 AI-assistable moments, then delivered a Next.js and Neon training platform with four courses and 16 modules for 34 trainees.

Freelance Web Developer

Oct 2024 – Present

Richard Álvarez | Sole Proprietorship

Chicago, IL

- Design and ship responsive web applications for creative and commercial clients using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, and object storage.
- Delivered two public-facing websites and continue developing several private web projects.
- Built Python AI projects, including Billy, a paper-trading LLM agent, and Half-Time, a soccer prediction model.

IT Assistant

Sep 2023 – Feb 2024

Kenyon College Library & Information Services

Gambier, OH

- Supported workstation imaging, hardware deployment, printer and AV troubleshooting, and secure drive destruction, and built a spreadsheet tool to automate classroom maintenance scheduling.

Education

Kenyon College

Aug 2020 – May 2024

Bachelor of Arts in Film

Gambier, OH

Minor in History; Concentration in Integrated Program in Humane Studies (IPHS)

Courses: Software Development, Data Structures and Programming Design, AI for the Humanities

Technical Skills

AI-Assisted Development: Claude Code, Codex CLI, GitHub Copilot, multi-agent coding workflows, spec-driven briefs, reviewing and testing AI-generated code

Agent Systems: MCP, tool calling, structured outputs with Zod and Pydantic, RAG with embeddings and BM25, context and memory design, agent orchestration, LLM-as-judge monitoring, Anthropic, OpenAI, and Gemini APIs

Workflow and Integration: n8n, REST APIs, webhooks, Autotask, Microsoft Graph and Teams, identity and permissions, human approval controls

Reliability and Delivery: Docker, GitHub Actions, Linux and VPS, CI/CD, automated testing, validation, logging, audit trails, observability, incident response, LLM cost and budget controls

Product and Data: TypeScript, Python, Bun and Node.js, Hono and FastAPI, React and Next.js, PostgreSQL, SQLite FTS5, pandas, scikit-learn, XGBoost

Systems and Media: C++20, CMake, FFmpeg, PySceneDetect